

Preface: Scope, Validity, and Epistemological Position

This work does not propose a theory intended to be classified as “true” or “false” in an absolute sense. Such a dichotomy, while rhetorically common, does not reflect how scientific theories operate in practice. Scientific models are evaluated by their **domain of validity**: the range of scales, contexts, and questions for which they provide stable, predictive, and operationally useful descriptions.

The history of science makes this explicit. Large-scale engineering and urban planning routinely treat the Earth as locally flat, without implying ignorance of its global curvature. Newtonian mechanics was not invalidated by relativity; it was **embedded** as a local approximation within a broader framework. A theory that were “false” in an absolute sense could not remain operationally effective in any regime.

From this perspective, the aim of the present work is neither to refute nor to replace established evolutionary models—most notably classical Darwinian frameworks—but to **clarify their scope** and to introduce a complementary theoretical layer addressing phenomena that lie outside their original domain.

Darwinian theory provides a powerful account of population-level dynamics driven by variation, selection, and differential reproduction. However, it was not formulated as a theory of **functional invariants**, nor as a theory of the persistence of necessary biological functions across changing material substrates. When pressed into this role, it can generate apparent paradoxes or misleading narratives, such as the assumption that the loss of a specific genetic structure necessarily implies the loss of the biological function it historically supported.

The Theory of Axiomatic Necessity (TNA) is introduced precisely at this level. It does not compete with evolutionary dynamics, but rather operates as a **higher-order structural framework**, concerned with the conditions under which certain functions must persist in order for a system to remain viable. In this sense, TNA stands in a relation to Darwinian theory analogous to that between thermodynamics and statistical mechanics, or between information theory and specific physical implementations of communication channels.

Throughout the paper, analogical language (e.g., hardware/software, error accumulation, discharge capacity, saturation) is employed as a **formal heuristic**, not as a literal reduction. These constructs are used to describe shared structural properties of complex systems—such as error correction, redundancy, collapse of support, and functional migration—independently of

the particular material realization. Their legitimacy rests on relational coherence and explanatory power, not on ontological identity.

This work explicitly does not claim to exhaustively describe biological reality. Its objective is narrower and more precise: to provide a coherent framework for understanding why certain functional states reappear, persist, or reconfigure over evolutionary time, even as the specific genetic or structural supports that implement them change or disappear.

Accordingly, critiques framed in terms of absolute truth or falsity are misplaced. The appropriate criterion is instead:

Within which domain does this framework offer explanatory or predictive advantages, and which classes of problems does it render tractable that remain unresolved or ill-posed under existing models?

It is under this criterion that the arguments presented in the following sections should be evaluated.